USV MPC Controller Optimization: Technical Defense & Parameter Breakdown

Autonomous Surface Vehicle (USV) Control System • Comparative Analysis: PID vs. Conventional MPC vs. Battery-Aware MPC vs. ML-Tuned MPC

Key Takeaway for Professor / Guide:
*Earlier baseline MPC failed to outperform PID because it was mathematically blind to lateral path displacement (cross-track error e_y was missing from the state vector), suffered from thrust saturation overshooting waypoints, and lacked dynamic weight adaptation. By adding e_y into the state-space model, calibrating thrust limits, and introducing Neural Network dynamic gain scheduling, MPC achieved **75.2% lower tracking error** and **46.5 s faster transit time** than PID.*

1. Core Technical Defense (Answering 'Why Couldn't MPC Get Good Results Before?')

If your professor asks: *'Why did MPC perform worse than PID earlier, and why couldn't you achieve superior performance initially?'*, use the following 4 scientific reasons:

Issue 1: State-Space Incompleteness (The 'Blind' Controller Problem)

- *Previous Defect:* The initial MPC model defined an error state vector of only 3 states: x = [e_u, r, e_psi]^T. Cross-track displacement error (e_y) was completely missing from the state vector and the quadratic cost matrix Q.
- *Result:* The QP solver minimized heading error and speed error, but was completely blind to lateral drift off the path caused by wind and currents. The USV accumulated 6–7+ meters of cross-track deviation.
- *Fix:* We expanded the state-space formulation to 4 states: x = [e_u, r, e_psi, e_y]^T, incorporating the kinematic cross-track coupling equation de_y/dt = u_ref * e_psi into the discretized system matrix A_d.

Issue 2: Thruster Saturation & High-Inertia Corner Overshoot

- *Previous Defect:* The thrust limit was set to T_max = 300.0 N. In an effort to minimize surge error, the baseline controller demanded maximum peak thrust (over 1300 W), accelerating the vessel into sharp waypoint turns at excessive speed.
- *Result:* Because the vessel possesses significant yaw inertia and hydrodynamic damping while the differential yaw moment is physically capped at 45 N·m, the rudder torque could not rotate the hull fast enough. The USV skidded and overshot corners, taking >1200 seconds or timing out.
- *Fix:* Calibrated T_max = 150.0 N to match achievable turning rates, eliminating thruster saturation and swerving.

Issue 3: Drag Linearization Deficit in Disturbance Vector

- *Previous Defect:* The affine disturbance offset vector D_d omitted the steady-state hydrodynamic drag term a_u * u_ref.
- *Result:* The linear prediction model underestimated hydrodynamic resistance, leading to persistent steady-state surge speed droop.
- *Fix:* Corrected disturbance offset to D_d = [(d_u + a_u * u_ref) * Ts; 0; 0; 0], ensuring zero-bias speed tracking.

Issue 4: Inherent Conflict in Static Cost Weights (Resolved via ANN)

- *Previous Defect:* In standard MPC, weight matrices Q and R are constant. High surge weight prevents slowing down for turns; low surge weight slows the entire voyage.

• *Result:* Fixed weights cannot dynamically trade off tight waypoint navigation against high-speed straight-line cruising.
• *Fix:* Integrated an Artificial Neural Network (ANN) that dynamically adapts cost weights in real time: sharply de-rating surge priority and boosting cross-track stiffness (w_cte) when entering turns.

2. Complete Parameter & Code Modifications Summary

The exact parameters and code structures modified across the codebase:

Parameter / Component	Original Baseline	Optimized Setting	File Location	Technical Objective
Error State Vector x	3 States: [e_u; r; e_psi]	**4 States:** **[e_u; r; e_psi; e_y]**	conventional_MPC.m battery_aware_MPC.m	Incorporates direct lateral trajectory position error into QP solver.
Coupling Matrix A_d	3x3 Matrix (No CTE dynamics)	**4x4 Matrix** A_d(4,3) = u_ref * Ts A_d(4,4) = 1.0	conventional_MPC.m battery_aware_MPC.m	Models cross-track drift rate as function of heading deviation.
Max Thrust (T_max)	300.0 N	**150.0 N**	MPC_parameters.m	Eliminates bang-bang saturation and excessive cornering speed.
Cross-Track Weight (w_cte)	0.0 (Unpenalized)	**10.0 (Static)** **120 to 870 (ML)**	MPC_parameters.m train_neural_network.m	Imposes direct quadratic penalty on distance away from centerline.
Heading Weight (w_psi)	20.0	**40.0 (Static)** **400 to 1342 (ML)**	MPC_parameters.m train_neural_network.m	Stiffens yaw tracking for rapid bow realignment.
Disturbance Vector D_d	[d_u * Ts; 0; 0]	**[(d_u + a_u*u_ref)*Ts; 0; 0; 0]**	conventional_MPC.m battery_aware_MPC.m	Compensates steady-state drag to eliminate velocity sag.
ML Speed De-rating (w_spd)	Constant ~100.0	**max(20, 100 - 15*|e_y|)**	train_neural_network.m	Intelligently slows down vessel during turns to allow clean rotation.
ML CTE Target (w_cte)	Static / Flat	**120 + 150*min(5, |e_y|)**	train_neural_network.m	Scales recovery priority aggressively with lateral displacement.

3. Empirical Verification: Simulation Results Benchmark

Quantitative results from the full simulation runs across all 6 benchmark models (from ml_comparison_results.csv):

Controller Architecture	Travel Time (s)	Cross-Track RMSE (m)	Energy (Wh)	Final SOC (%)	Improvement vs PID
PID Baseline	1021.8 s	3.789 m	158.4 Wh	66.80 %	Baseline Reference

Controller Architecture	Travel Time (s)	Cross-Track RMSE (m)	Energy (Wh)	Final SOC (%)	Improvement vs PID
PID + ML (ANN)	996.9 s	3.119 m	134.2 Wh	68.82 %	17.7% Error Reduction
PID + ML + Battery	1009.5 s	3.152 m	127.6 Wh	69.36 %	16.8% Error Reduction
Conventional MPC	1003.2 s	1.309 m	143.9 Wh	68.00 %	**65.5% Error Reduction**
Battery-Aware MPC	995.4 s	1.262 m	142.4 Wh	68.14 %	**66.7% Error Reduction**
MPC + ML (ANN)	978.0 s	0.978 m	134.5 Wh	68.79 %	**74.2% Error Reduction**
MPC + ML + Battery (BEST)	**975.3 s**	**0.938 m**	**133.0 Wh**	**68.91 %**	**75.2% Error Reduction (Top)**

4. Quick Presentation Script for Viva / Professor Meeting

30-Second Summary to Speak in Meeting:

- *'Sir, previously our MPC was constrained by a 3-state error model that lacked cross-track displacement, and its static thrust limit was set too high, causing overshoot during cornering.'*
- *'We resolved this by expanding the model to 4 states—explicitly including cross-track error e_y and kinematic drift coupling—and calibrating the thrust ceiling to 150 N.'*
- *'Furthermore, we introduced an Artificial Neural Network that dynamically de-rates speed during sharp turns and sharply scales path-stiffness weights. As a result, MPC+ML+Battery achieved 0.938 m tracking accuracy—75% better than PID—while arriving 46.5 seconds faster and saving 16% battery energy.'*